

Einführung in die Kern-, Teilchen und Astrophysik

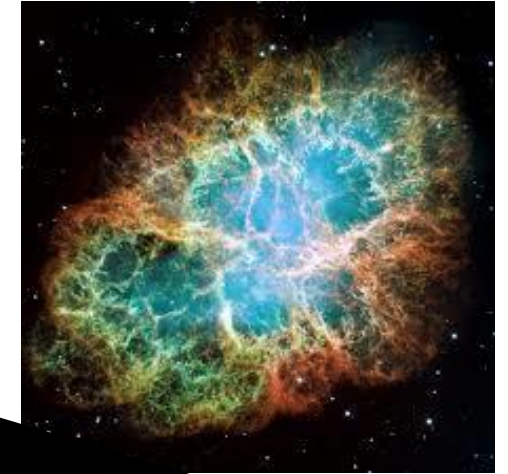
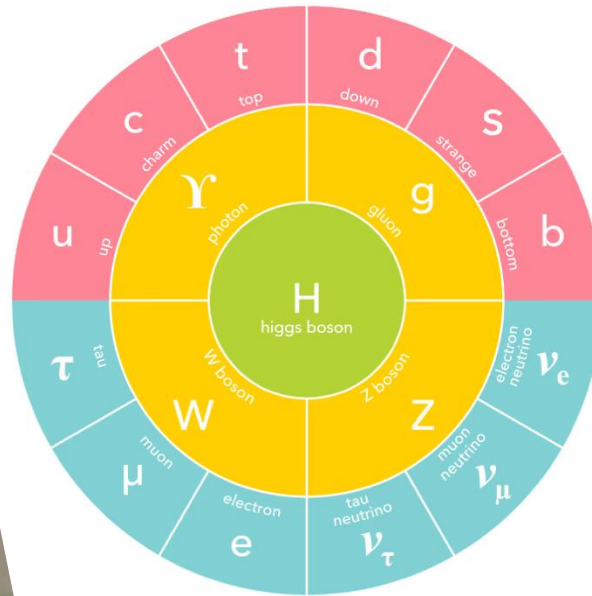
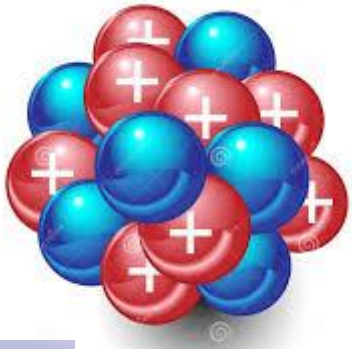


Table of contents

Lecture	Date	Theme	Topic	Sub-Topics
1	14/10/25	Introduction	Little History of particle physics	atoms, elementary particles
2	21/10/25	Particle Physics	Standard Model	Fermions, Bosons, Interactions,
3	28/10/25		Fundamental symmetries	Conservations, Feynman, Propagator,
4	04/11/25		Fundamental symmetries	Discrete symmetries; Cross section
5	11/11/25		Interaction with Matter	Bethe Bloch, gamma interaction
6	18/11/25		Particle detection	Gamma interactions, detector basics
7	25/11/25	Astrophysics	Detectors, accelerators	Detector types, accelerators, cyclotron Supernovae, cosmic rays, Supernovae, cosmic rays
8	02/12/25	Nuclear physics	Basics	Binding energies, Weizäcker
9	09/12/25		Nuclear reactions	Beta, alpha, fission
10	16/12/25		Nuclear reactors	Radiocarbon, gamma transitions
11	13/01/26	Particle / Nuclear	High energy physics 1	Deep inelastic scattering
12	20/01/26		High energy physics 2	QCD
13	27/01/26	Particle Physics	Weak interactions	
14	03/02/26	Astrophysics	Neutrinos	

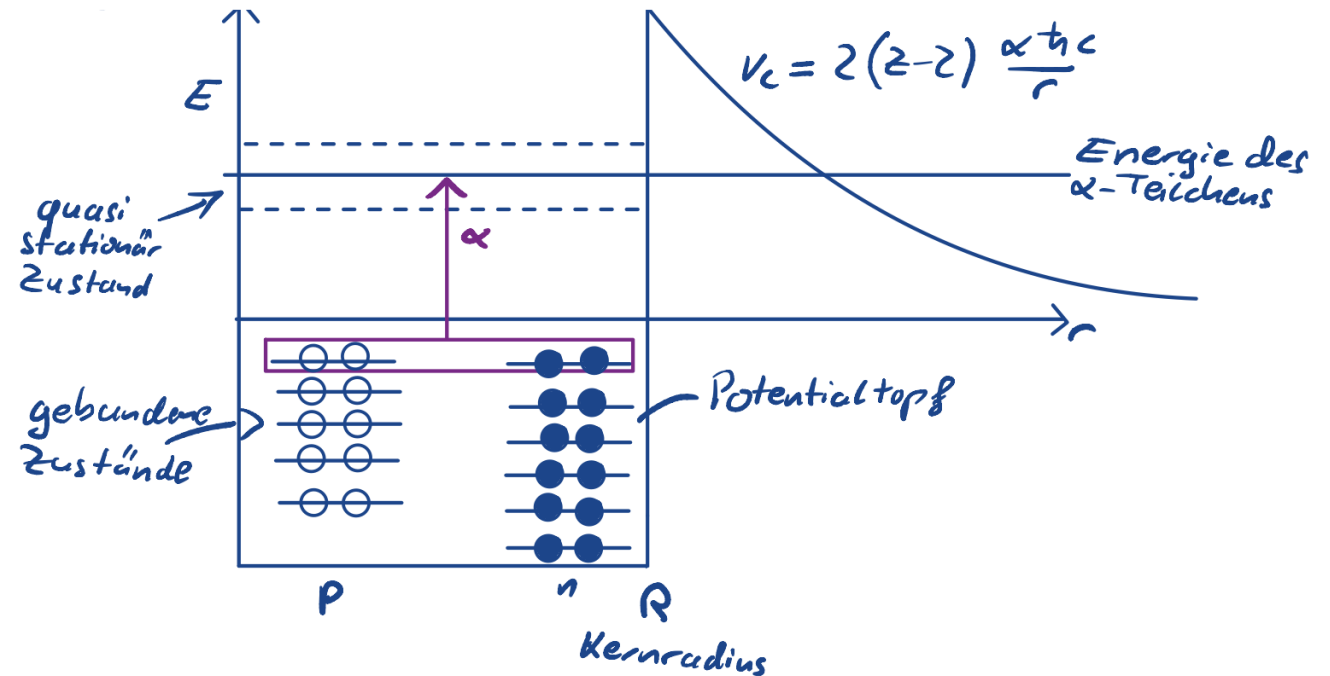
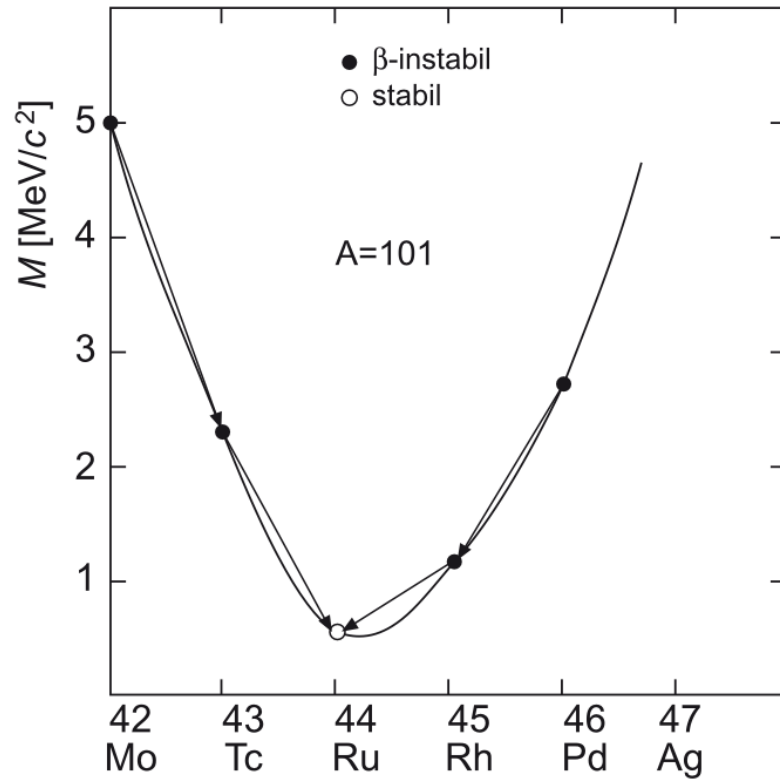


Quiz am 23.12.25 --- Online in Moodle

- Bestandenes Quiz als Voraussetzung für Notenbonus
- Komplett online um 14:30 Uhr am Dienstag 23.12.25
- Länge: 20-25 min
- Test Quiz online am Freitag
- Zugang für beide Quiz in moodle
- Email mit Details wird folgen!



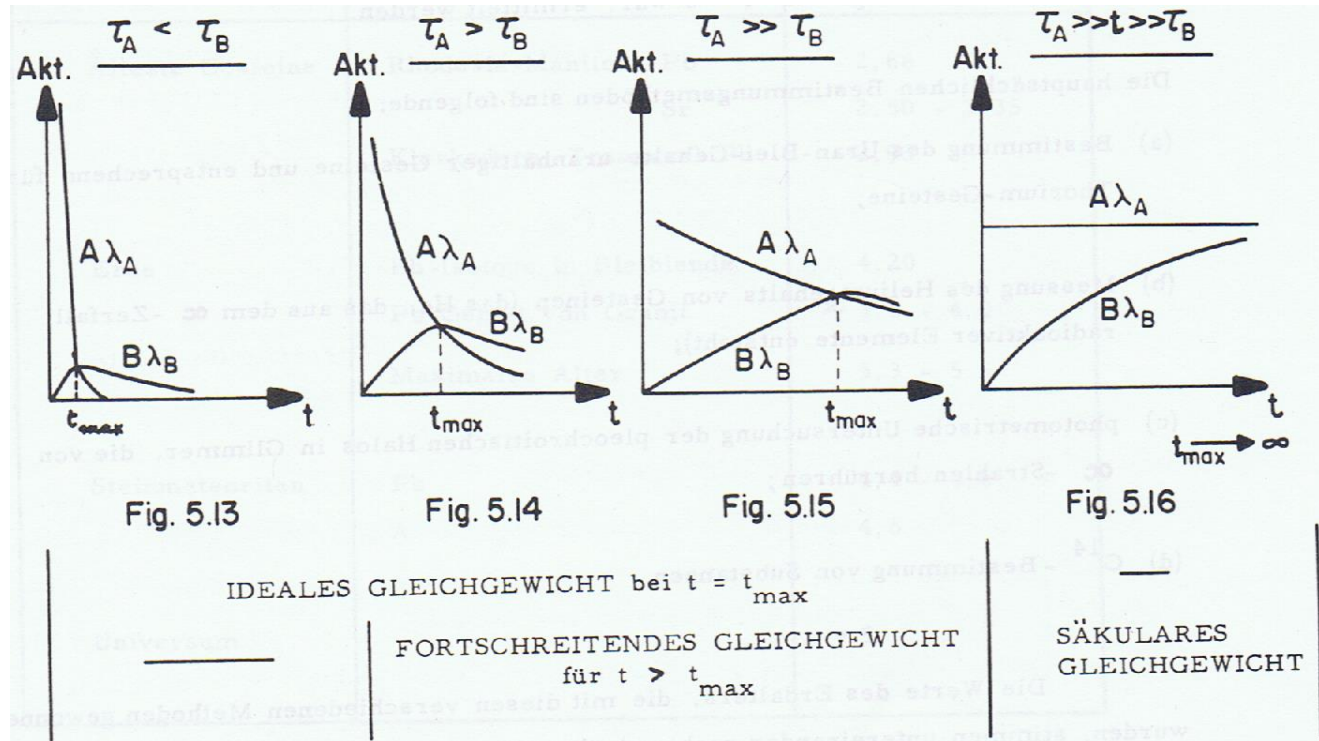
Zusammenfassung der letzten Vorlesung



Bindungsenergien und
Betazerfälle

Alpha Zerfall

Kernspaltung



Aktivität von B unter 4 verschiedenen Fällen.

Das „ideale Gleichgewicht“ erreicht man für $t_{\max}$ wenn die Aktivität maximal wird (berechenbar aus Kurvendiskussion obiger Formel).

Für $\tau_A \gg \tau_B$ gleichen sich die Aktivitäten von Mutter- und Tochterkern an.

Man spricht vom „säkularen Gleichgewicht“, wenn auch noch die Messzeit $t \gg \tau_B$

Uran- und Thorium – Zerfallsketten

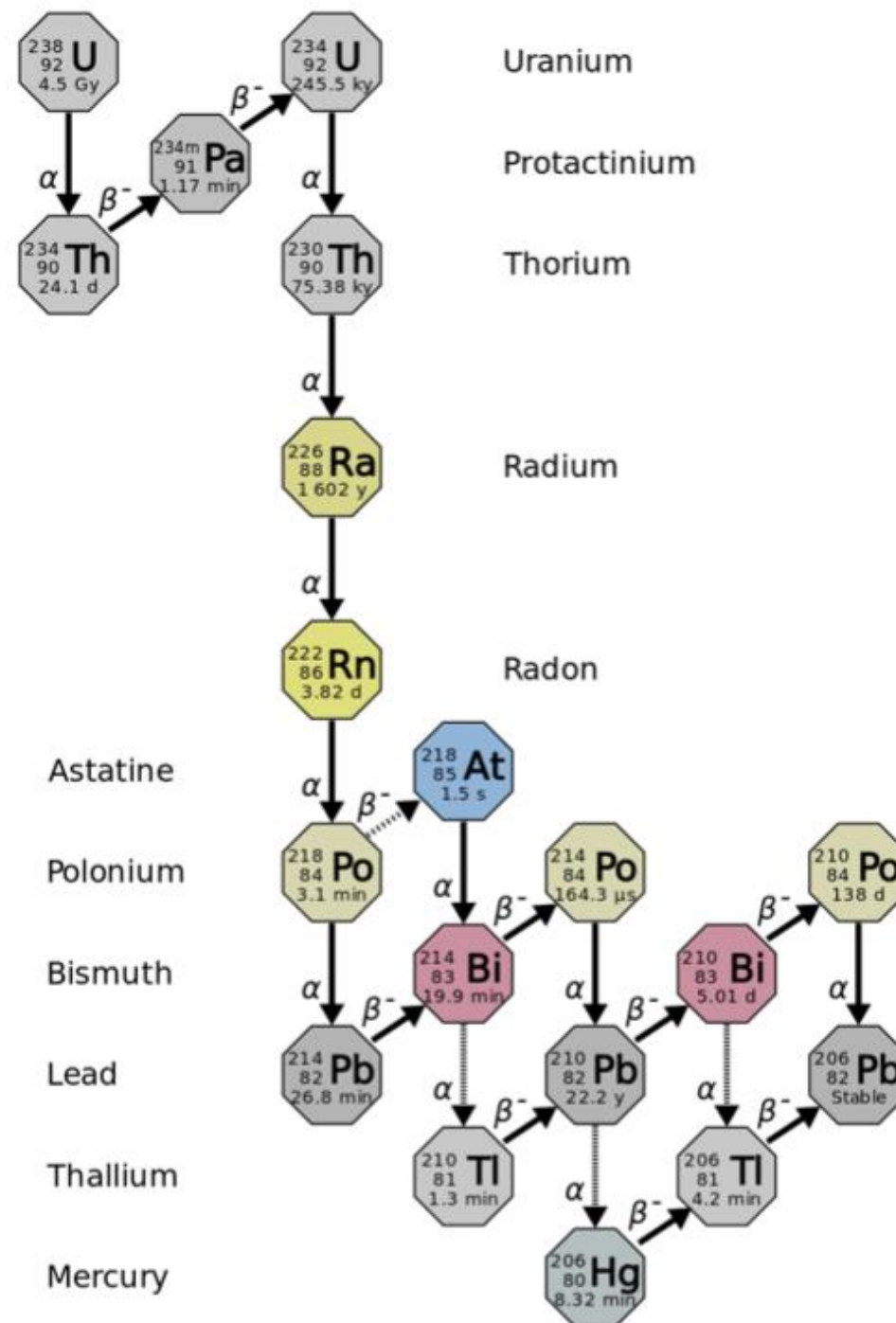
Gezeigt wird die Kette von ^{238}U

Np 229 4.0 m	Np 230 4.6 m	Np 231 48.8 m	Np 232 14.7 m	Np 233 36.2 m	Np 234 4.4 d	Np 235 396.1 d	Np 236 2.117 d	Np 237 2.355 d
α 5.690	α 6.66	α 6.248 3771; 684; 284...	ϵ 327; 820; 867; 664; 282...	ϵ 5.54 3012; 299; 547...	ϵ 1.75; 1559; 1528; 1602; 370; 100	ϵ 5.025; 5.007; 1559; 1528; 1602; 370; 100	ϵ 1.75; 1559; 1528; 1602; 370; 100	ϵ 1.75; 1559; 1528; 1602; 370; 100
U 228 9.1 m	U 229 58 m	U 230 20.8 d	U 231 4.2 d	U 232 68.9 a	U 233 1.592 · 10 ⁵ a	U 234 0.0054 a	U 235 0.7204 a	U 236 2.342 · 10 ⁷ a
α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...	α 6.68; 6.59...
Pa 227 38.3 m	Pa 228 22 h	Pa 229 1.50 d	Pa 230 17.4 d	Pa 231 3.276 · 10 ⁴ a	Pa 232 1.31 a	Pa 233 27.0 d	Pa 234 11.5 d	Pa 235 6.70 h
α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...	α 6.456; 6.416...
Th 226 31 m	Th 227 18.72 d	Th 228 1.913 a	Th 229 7880 a	Th 230 7.54 · 10 ⁴ a	Th 231 25.5 h	Th 232 100 a	Th 233 22.3 m	Th 234 24.10 d
α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...	α 6.336; 6.230...
Ac 225 10.0 d	Ac 226 29 h	Ac 227 21.773 a	Ac 228 6.13 h	Ac 229 62.7 m	Ac 230 122 s	Ac 231 7.5 m	Ac 232 119 s	Ac 233 145 s
α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14	α 5.830; 5.793; 5.732; 0.14

146

Ac 215 0.17 s	Ac 216 0.44 ms	Ac 217 0.74 μs	Ac 218 1.1 μs	Ac 219 11.8 μs	Ac 220 26 ms	Ac 221 52 ms	Ac 222 63 s	Ac 223 2.10 m	Ac 224 2.9 h	Ac 225 10.0 d	Ac 226 29 h	Ac 227 21.773 a	Ac 228 6.13 h
Ra 214 2.46 s	Ra 215 1.67 ms	Ra 216 2.0 ns	Ra 217 1.6 μs	Ra 218 25.6 μs	Ra 219 10 ms	Ra 220 23 ms	Ra 221 28 s	Ra 222 38 s	Ra 223 11.43 d	Ra 224 3.66 d	Ra 225 14.8 d	Ra 226 1600 a	Ra 227 42.2 m
Fr 213 34.6 s	Fr 214 3.35 ms	Fr 215 0.09 μs	Fr 216 0.70 μs	Fr 217 16 μs	Fr 218 22 ms	Fr 219 21 ms	Fr 220 27.4 s	Fr 221 4.9 m	Fr 222 14.2 m	Fr 223 21.8 m	Fr 224 3.3 m	Fr 225 4.0 m	Fr 226 48 s
Rn 212 24 m	Rn 213 19.5 ms	Rn 214 6.5 ns	Rn 215 2.3 μs	Rn 216 45 μs	Rn 217 0.54 ms	Rn 218 35 ms	Rn 219 3.96 s	Rn 220 55.6 s	Rn 221 25 m	Rn 222 3.825 d	Rn 223 23.2 m	Rn 224 1.78 h	Rn 225 4.5 m
At 211 7.22 h	At 212 119 ms	At 213 0.11 μs	At 214 0.76 μs	At 215 0.1 ms	At 216 0.3 ms	At 217 32.3 ms	At 218 ~2 s	At 219 0.9 m	At 220 3.71 m	At 221 2.3 m	At 222 54 s	At 223 50 s	
Po 210 138.38 d	Po 211 5.016 s	Po 212 45.1 s	Po 213 4.2 μs	Po 214 164 μs	Po 215 1.78 ms	Po 216 0.15 s	Po 217 1.53 s	Po 218 3.05 m	Po 219 300 ns	Po 220 >300 ns			
Pb 209 100 a	Pb 210 5.013 d	Pb 211 9.8 m	Pb 212 10.64 h	Pb 213 42.8 m	Pb 214 26.8 m	Pb 215 19.9 m	Pb 216 36.9 s	Pb 217 98.5 s	Pb 218 33 s				
Pb 208 52.4 a	Pb 209 3.253 h	Pb 210 22.3 a	Pb 211 36.1 m	Pb 212 10.64 h	Pb 213 10.2 m	Pb 214 26.8 m							
Tl 207	Tl 208	Tl 209	Tl 210	Tl 211	Tl 212								

Farbcode der
Nuklidkarte:
blau = β^- Zerfall
rot = β^+ Zerfall
gelb = α Zerfall



Uranium

Protactinium

Thorium

Radium

Radon

Astatine

Polonium

Bismuth

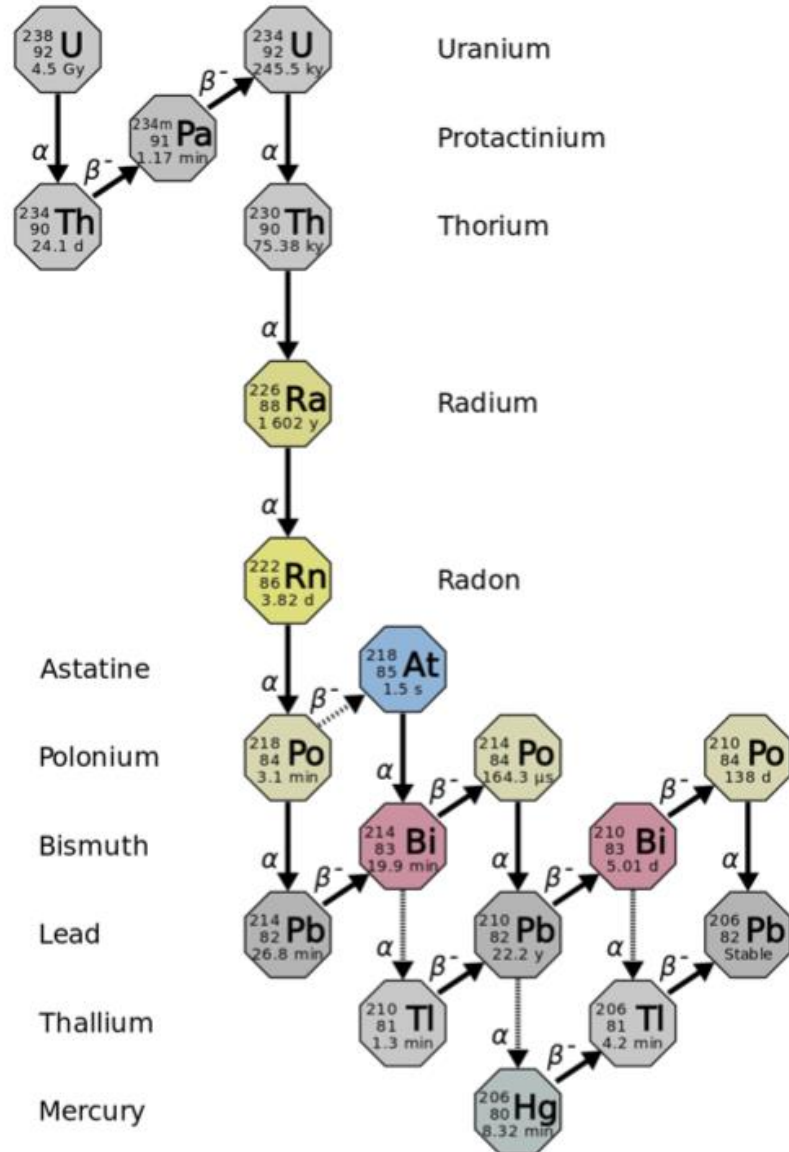
Lead

Thallium

Mercury

Beispiel aus der "Praxis"

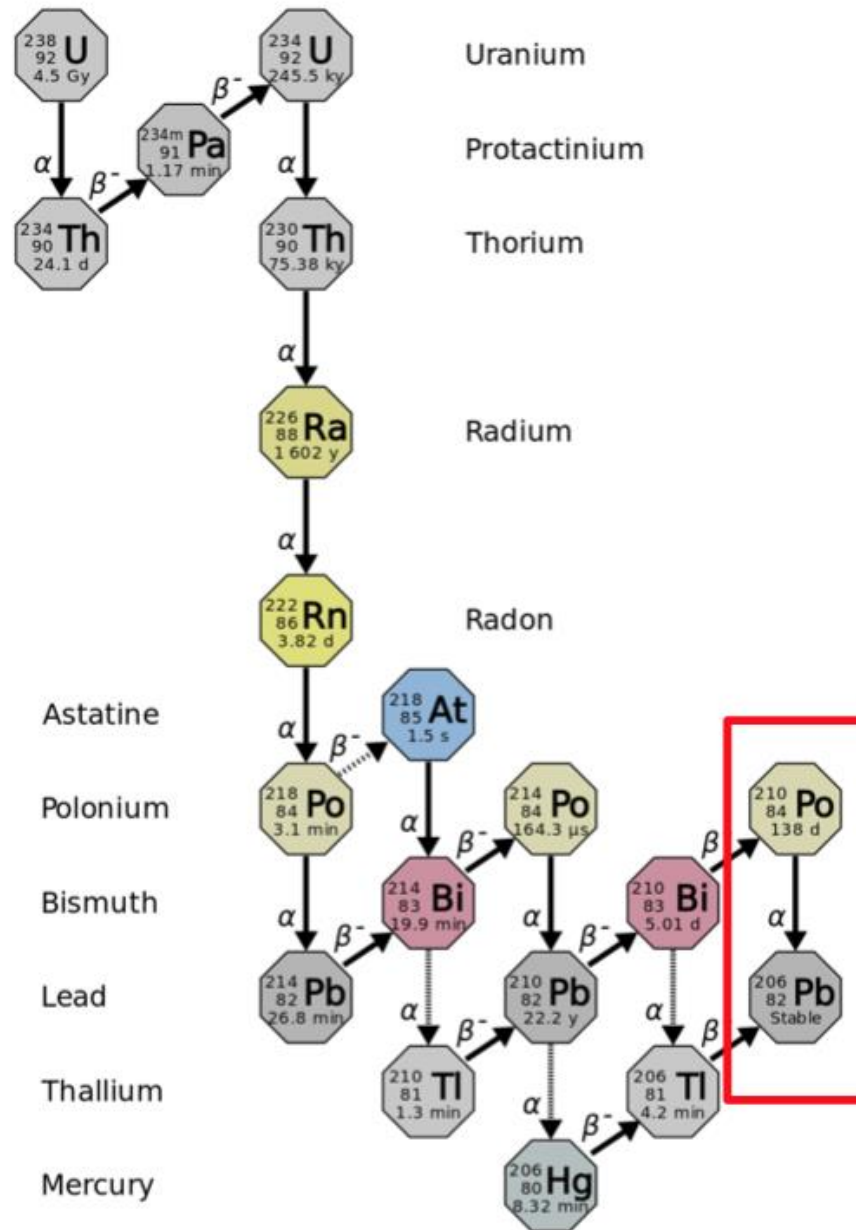
CRESST Experiment zur Suche nach Dunkler Materie



Beobachtung von **Störsignalen** (Untergrund),
welche von **Dunkler** Materie Signalen
ununterscheidbar sind!

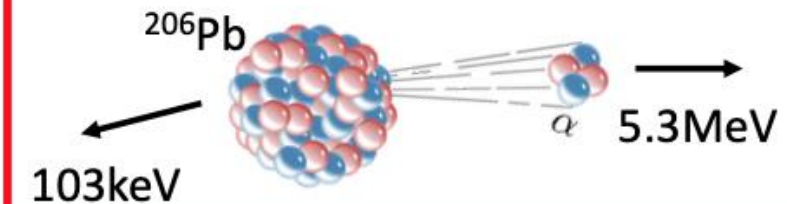
Beispiel aus der "Praxis"

CRESST Experiment zur Suche nach Dunkler Materie



Beobachtung von **Störsignalen** (Untergrund),
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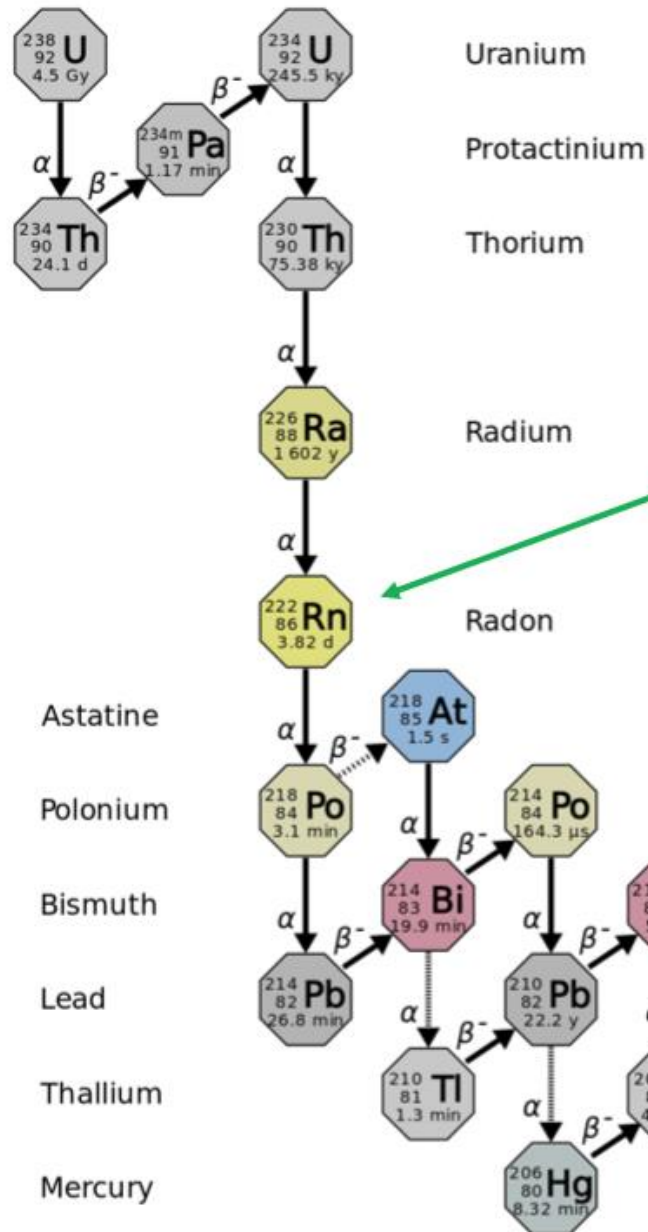
Identifizierung des Untergrundes: ^{210}Po (Alpha Strahler)



Beispiel aus der "Praxis"



Fragestellung: Ursprung der ^{210}Po Kontamination ??



Option 1:



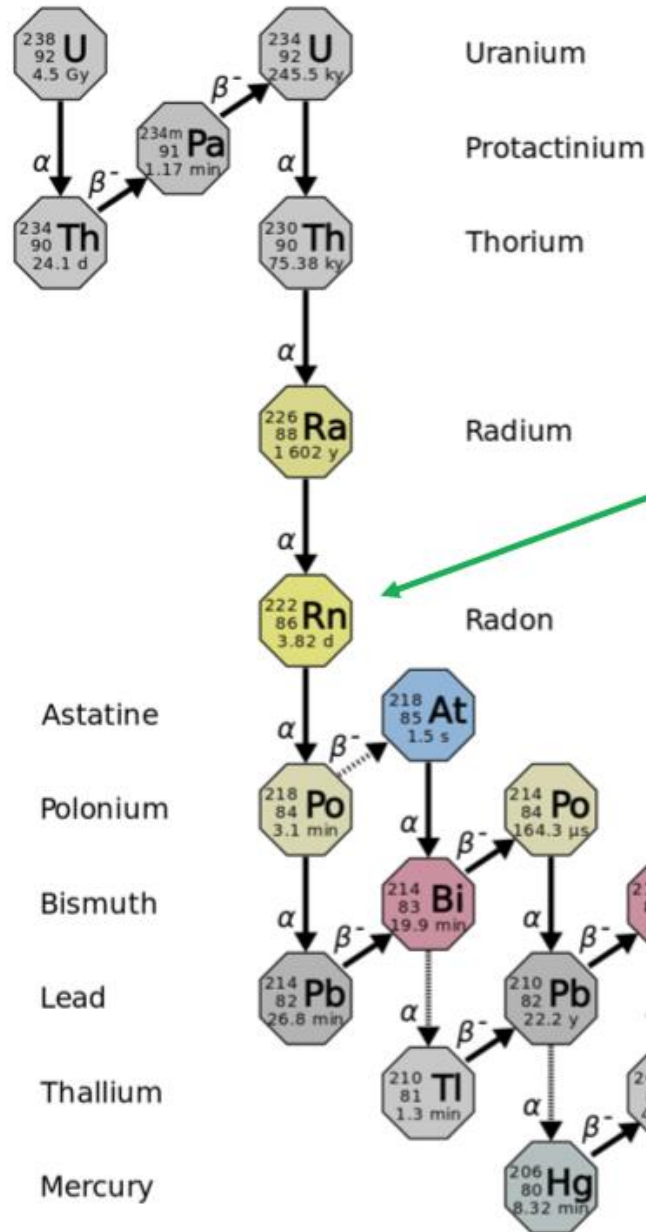
Option 2:



Beispiel aus der "Praxis"



Fragestellung: Ursprung der ^{210}Po Kontamination ??



Option 1:



Option 2:

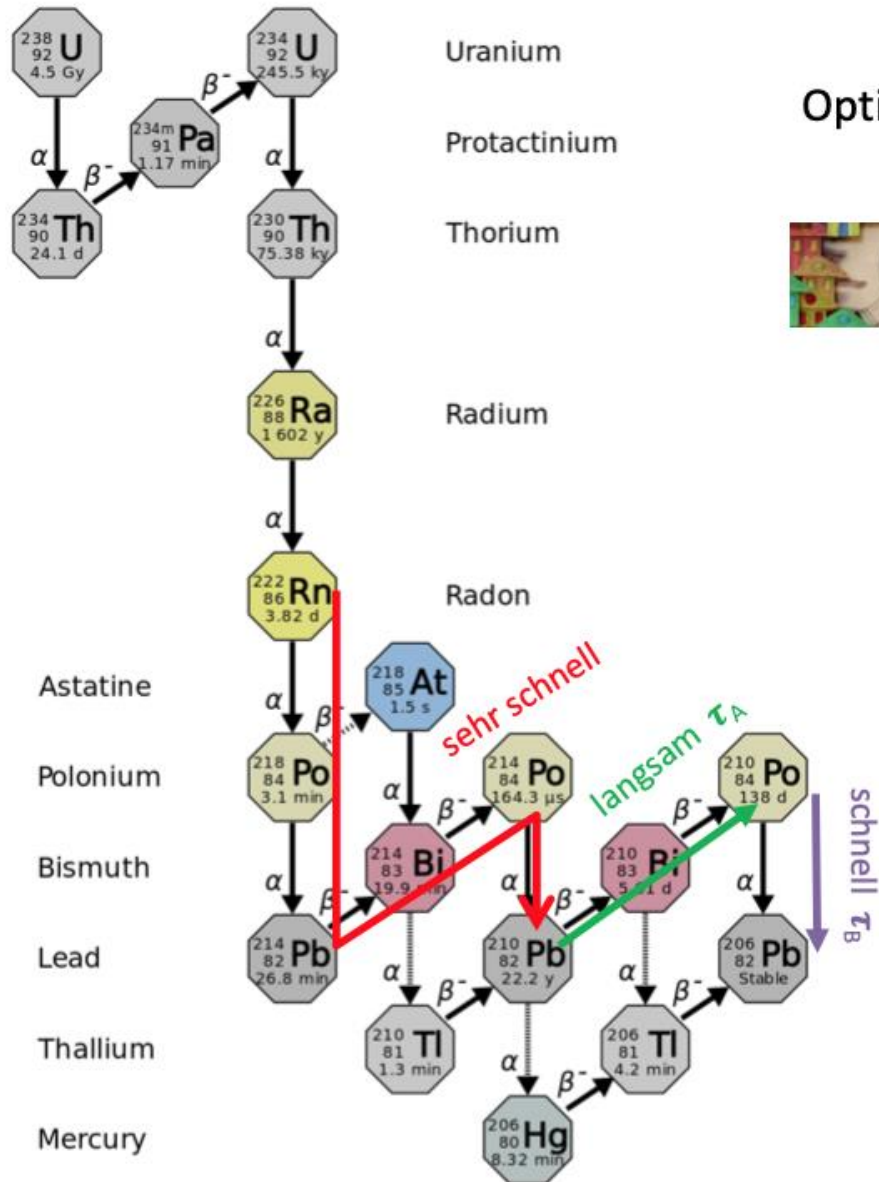


- Messzeit: mehrere Jahre
- ^{210}Po hat eine Halbwertszeit von 138 Tagen
- Durch zeitliche Entwicklung der ^{210}Po Aktivität kann man die Ursache der Strahlung bestimmen!

Welche zeitliche Entwicklung der ^{210}Po Alpha Aktivität erwarten sie für Option 1 bzw. 2 ?

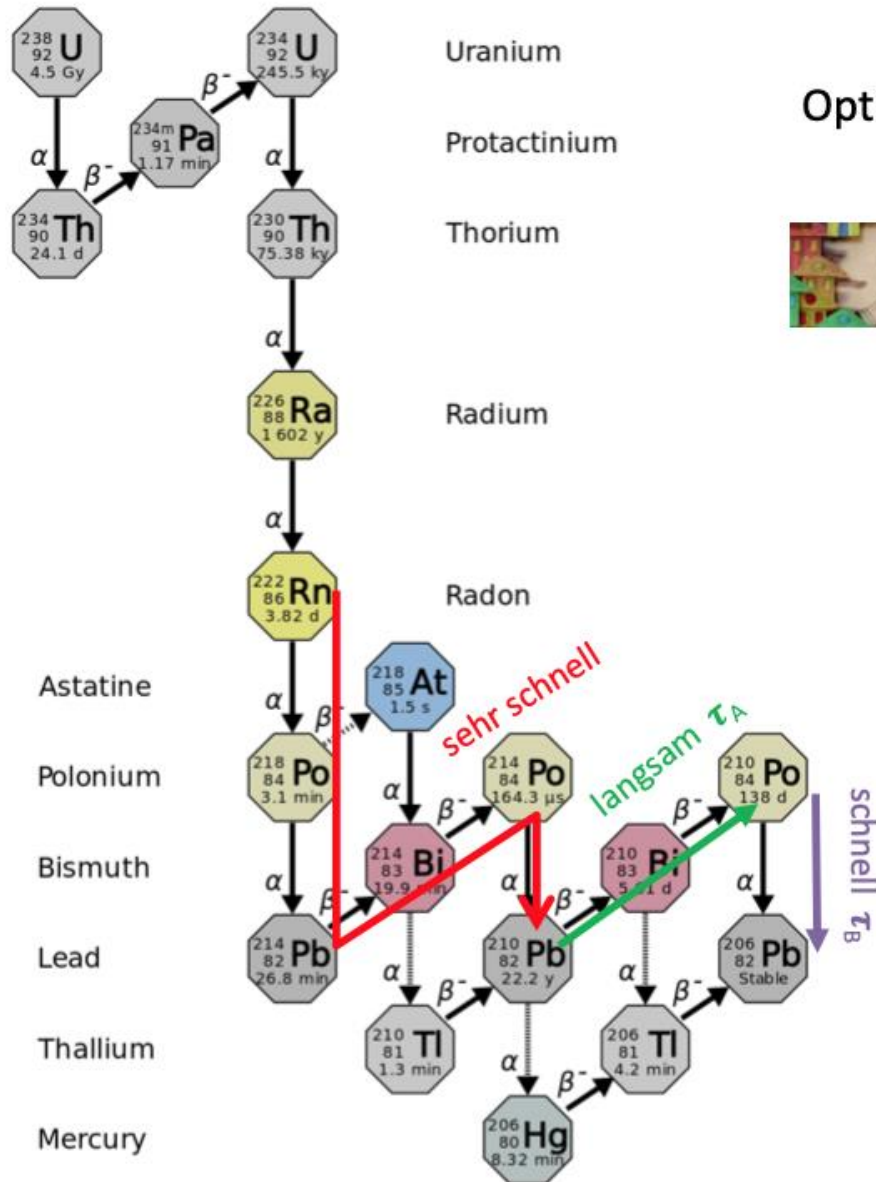
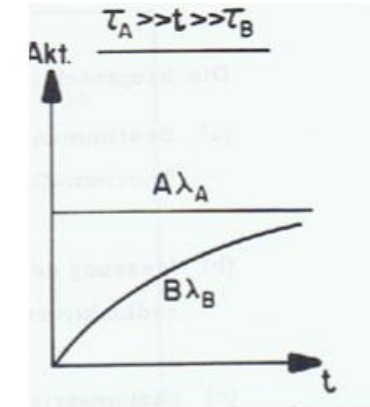
Lösung

Option 1:



Lösung

Option 1:

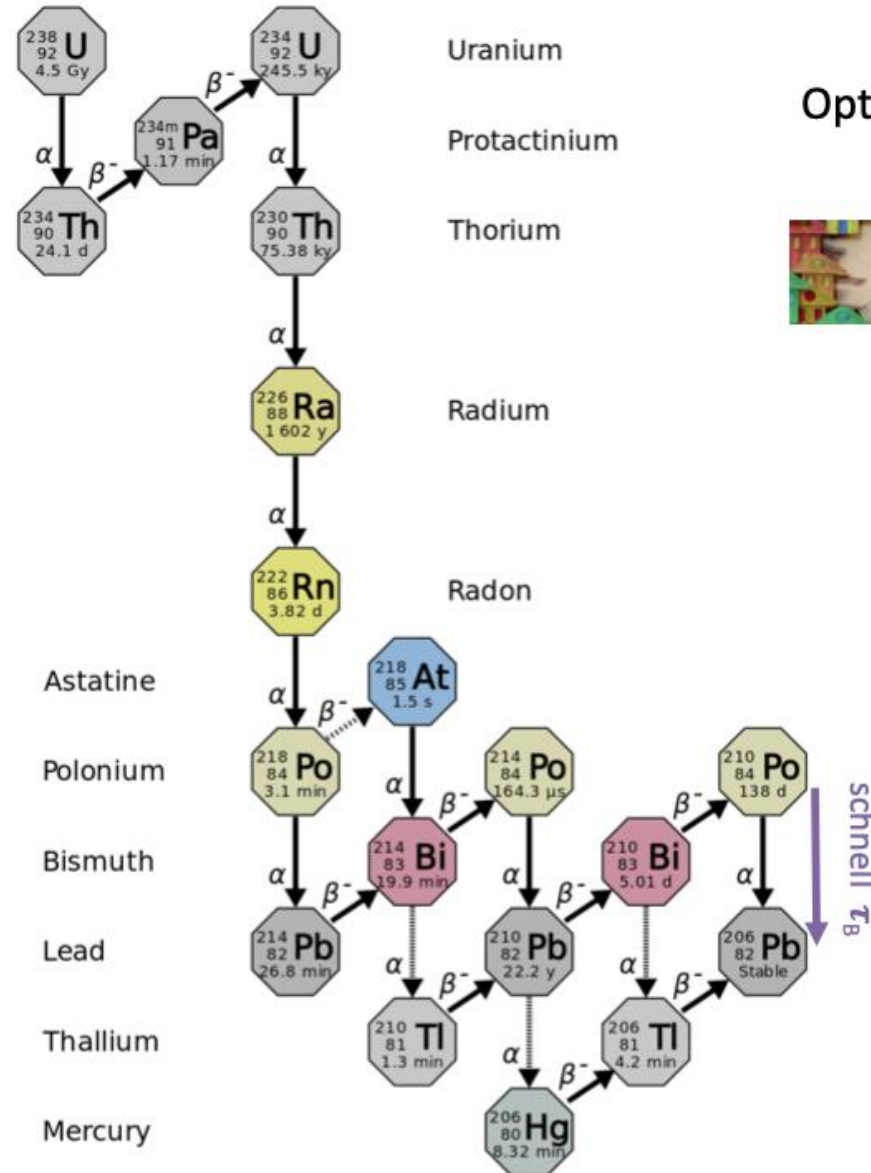


Lösung

Option 1:

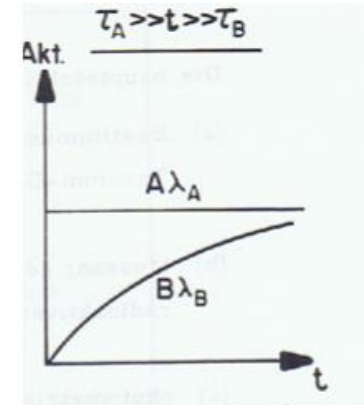


Option 2:

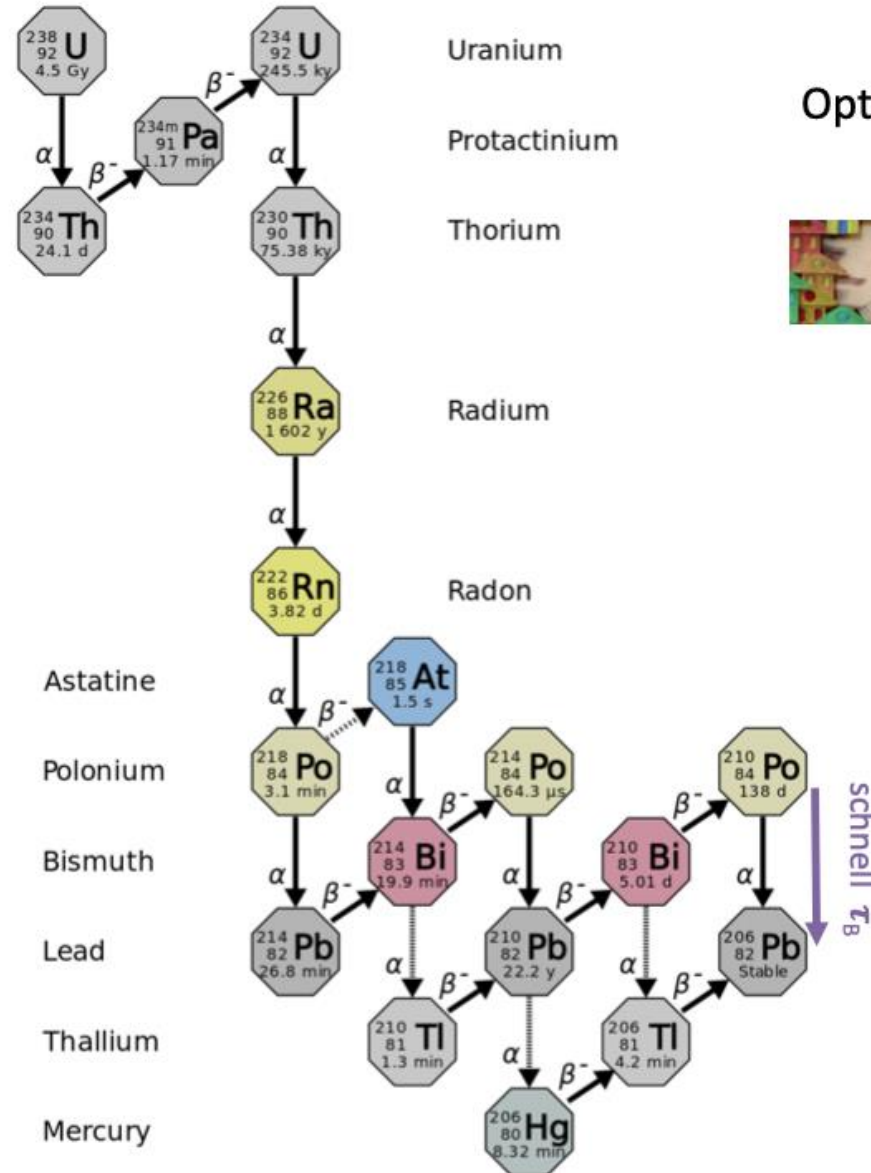
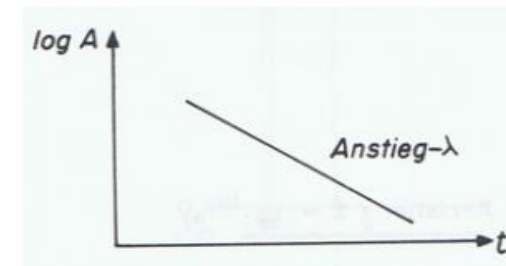


Lösung

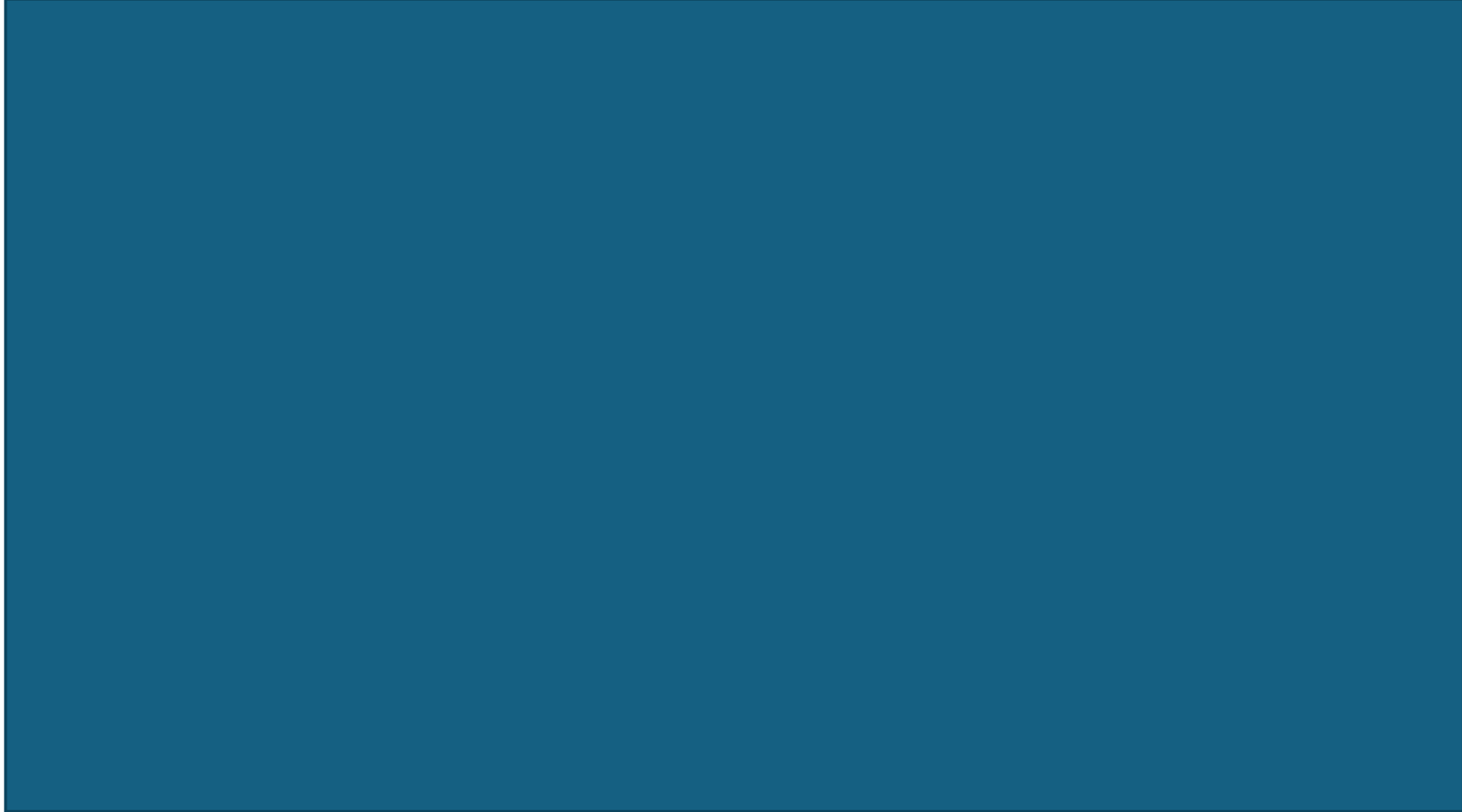
Option 1:



Option 2:

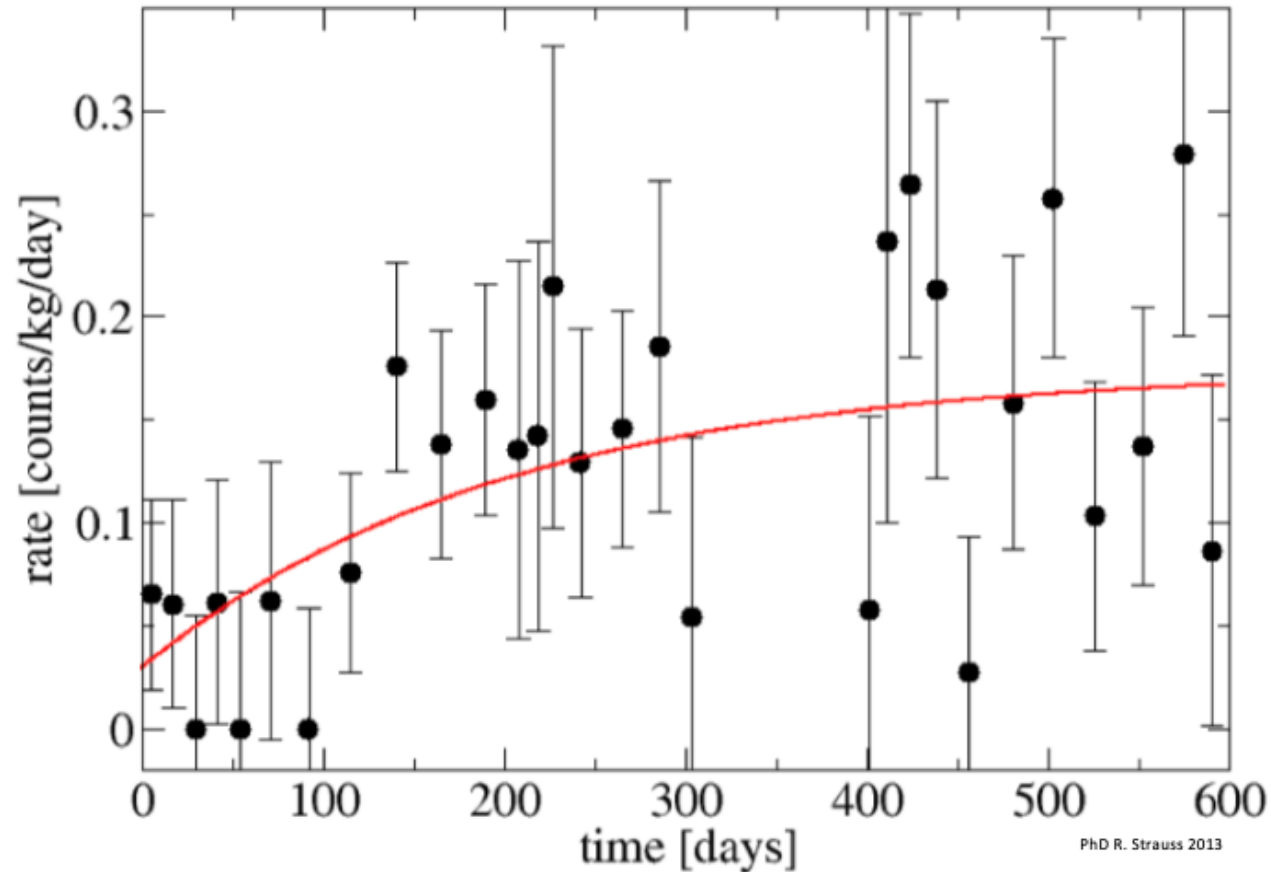
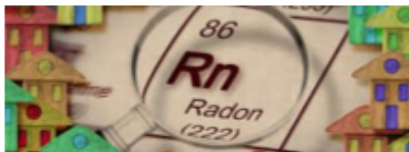


Lösung



Lösung

Rate von ^{206}Pb
Rückstößen
=
 ^{210}Po Aktivität



Der Ratenanstieg mit in etwa der Halbwertszeit von ^{210}Po deutet darauf hin, dass Rn (kurz) vor der Messung in den Detektoren implantiert wurde.

Anwendung **Altersbestimmung**

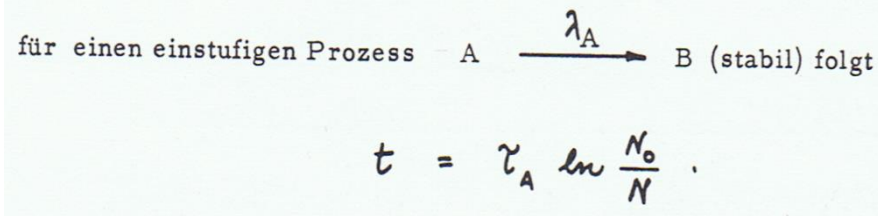
Die wichtigsten Methoden sind

- ^{14}C – Methode für organische Substanzen (z.B. Iceman, bzw. „Ötzi“ starb unterhalb der Finailspitze auf 3200m vor ca. 5250 Jahren). Halbwertszeit ^{14}C : 5730 a
- Uran/Thorium – Blei Gehalt von Gestein und Meteoriten
Halbwertszeiten ^{238}U : 4.5×10^9 a ^{232}Th : 1.4×10^{10} a
- He – Gehalt von Gesteinen und Meteoriten

Alter unseres Sonnensystems: $4,5 \times 10^9$ a

Vergleiche auch dazu das Vorkommen von ^{235}U mit nur 0,7%. Halbwertszeit: 7×10^8 a

Aus dem Zerfallsgesetz $N = N_0 e^{-\lambda_A t}$



Wenn A und B die Zahl der jeweiligen Kerne bezeichnen, gilt

$$t = \tau_A \ln \frac{A+B}{A}$$

Das Alter t ist dann bestimmbar, wenn man gute Annahmen für B zur Zeit $t=0$ machen kann und wenn es keine Konkurrenzprozesse gibt, die A und B sonst signifikant verändern

Radiokarbonmethode ^{14}C :

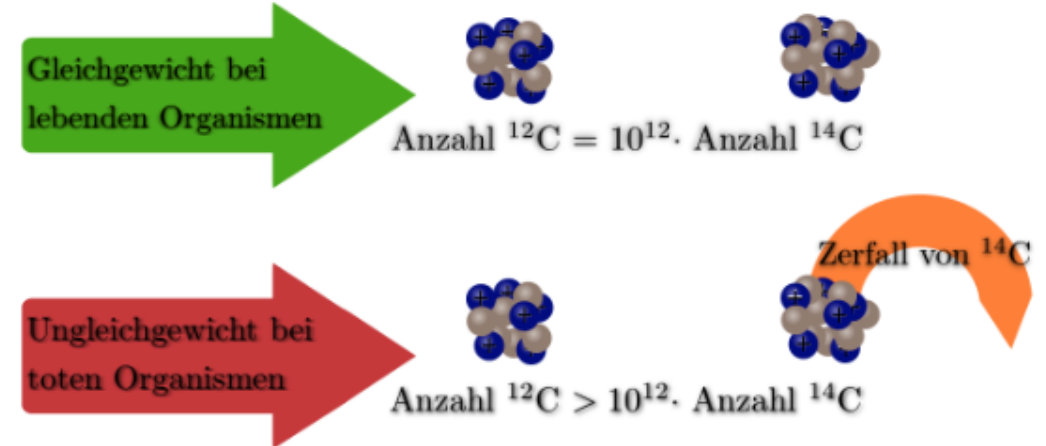
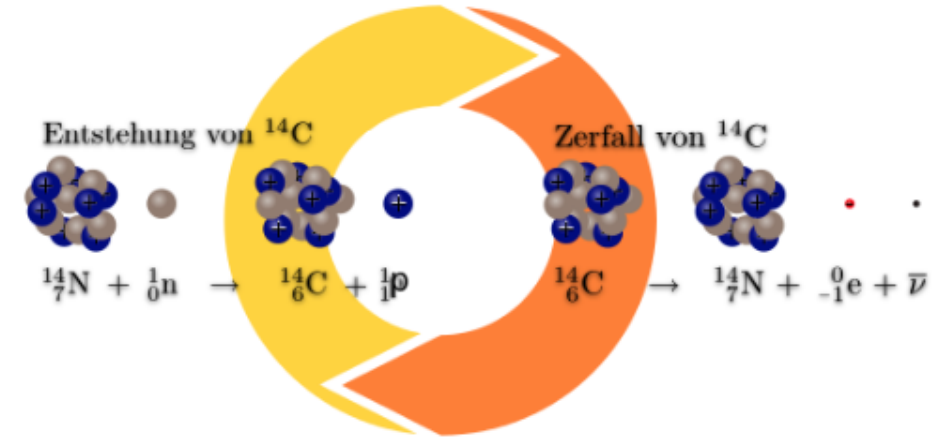
- Datierung von organischen Proben (300-50.000 Jahre)
- C-14 zerfällt mit Halbwertszeit von 5800 Jahren via Beta-Zerfall

Zerfallsgesetz:

$$\left(\frac{^{14}\text{C}}{^{12}\text{C}}\right) = \left(\frac{^{14}\text{C}}{^{12}\text{C}}\right)_{\text{Luft}} \cdot e^{-\lambda_{14}t} \quad \text{mit} \quad \lambda_{14} = 1,21 \cdot 10^{-4} \frac{1}{\text{a}}$$

Messmethoden:

- Zählrohr
- Flüssigszintillator
- Beschleuniger-Massenspektroskopie (AMS)



Bekannteste Anwendung:
Turiner Grabtuch

Reliquie von Jesus von Nazareth?

Radiokarbonmethode datiert Grabtuch auf
Mittelalter

Nature. Band 337, Nr. 6208, 16. Februar 1989, S. 611–615



Quelle: wikipedia

Bekannteste Anwendung: Turiner Grabtuch

Reliquie von Jesus von Nazareth?

Radiokarbonmethode datiert Grabtuch Mittelalter

Nature. Band 337, Nr. 6208, 16. Februar 1989, S. 61

Kuriosität: Verfälschung der Messergebnisse durch Auferstehung?

594

CORRESPONDENCE

NATURE VOL 337 16 FEBRUARY 1989

Shroud irradiated with neutrons?

Sir — If the shroud of Turin is in fact the burial cloth of Christ, contrary to its recent carbon-dated age of about 670 years (*Nature* 335, 663; 1988 and 337, 611; 1989), then according to the Bible it was present at a unique physical event: the resurrection of a dead body. Unfortunately, this event is not accessible to direct scientific scrutiny, but the image on the shroud, which still cannot be duplicated, appears to be a scorch, indicating that the body radiated light and/or heat. It may also have radiated neutrons, which would have irradiated the shroud and changed some of the nuclei to different isotopes by neutron capture. In particular, some ^{14}C could have been generated from ^{14}N . If we assume that the shroud is 1,950 years old and that the neutrons were emitted thermally, then an integrated flux of 2×10^{16} neutrons cm^{-2} would have converted enough ^{14}N to ^{14}C to give an apparent carbon-dated age of 670 years.

This flux of neutrons should have other measurable consequences. The neutron irradiation would probably not have been uniform, for example, so the $^{14}\text{C}/^{12}\text{C}$ ratio

should vary in different parts of the shroud. In addition, other unstable isotopes should have been formed. Several of these isotopes have half-lives long enough that they would still be present, yet short enough that they are not found naturally.

The unstable isotopes most likely to be found in the shroud are ^{36}Cl and ^{44}Ca . The presence of either would confirm that the shroud had been irradiated with neutrons. An accurate measurement of the ratio of either ^{36}Cl to ^{35}Cl or ^{44}Ca to ^{40}Ca (see table) would test the prediction of an integrated neutron flux of 2×10^{16} neutrons cm^{-2} . This may not be possible, however, because contamination with new sources of chlorine or calcium may have occurred from washings or other sources since the irradiation took place.

THOMAS J. PHILLIPS
High Energy Physics Laboratory,
Harvard University,
Cambridge, Massachusetts 02138, USA

HEDGES REPLIES—The processes suggested by Phillips were considered by the participating laboratories. However, for the reasons given below, the likelihood

Properties of selected parent/daughter isotopes

Parent isotope	Natural abundance (per cent)	Neutron capture cross-section (barns)	Daughter isotope	Half-life (years)	Predicted parent/daughter ratio
^{14}C	1.11	0.0009	^{14}C	5,730	1.2×10^{-10}
^{36}Cl	75.77	43	^{36}Cl	3.0×10^5	8×10^{-7}
^{44}Ca	96.94	0.40	^{44}Ca	1.0×10^5	8×10^{-9}

Predicted parent/daughter ratios (right-hand column) are for a new cloth irradiated with 2×10^{16} neutrons cm^{-2} . For $^{14}\text{C}/^{12}\text{C}$, the ratio is about 17 per cent higher than normal.

that they influenced the date in the way proposed is in my view so exceedingly remote that it beggars scientific credulity.

(1) No plausible physical mechanism has been proposed to explain how the resurrection was accompanied by a significant neutron flux. If a supernatural explanation is to be proposed, it seems pointless to make any scientific measurement on the shroud at all.

(2) Assuming a 'scientific' (but not yet articulated) explanation for the neutron flux, it is an amazing coincidence that the neutron dose should be so exactly appropriate to give the most likely date on historical grounds. (Arguably a total of 10^{16} neutrons (the number in a human body) would be available. Using Phillips' figures, this would be sufficient to impart a date of 100,000 years into the future. To produce a date within 100 years of the first recorded history of the shroud implies that the dose has been 'fine-tuned' to better than one part in a hundred million.)

(3) In fact, the dose proposed by Phillips is much too high, as he has not included the neutron capture by nitrogen in the cloth. A not untypical N content in linen is 1,000 p.p.m., for which a thermal neutron flux of $2 \times 10^{13} \text{ cm}^{-2}$ (that is, 1,000 times less) would be appropriate. This does not change the basic argument, but changes the chemical implications (even without recoil effects). The three dating laboratories used different types of chemical pre-treatments, (for example at Oxford we purified the cellulose), yet obtained equivalent results. This shows that any ^{14}C formed by neutron irradiation behaves chemically in the same way as the original ^{14}C . This is inherently unlikely because the original nitrogen is in a chemically quite